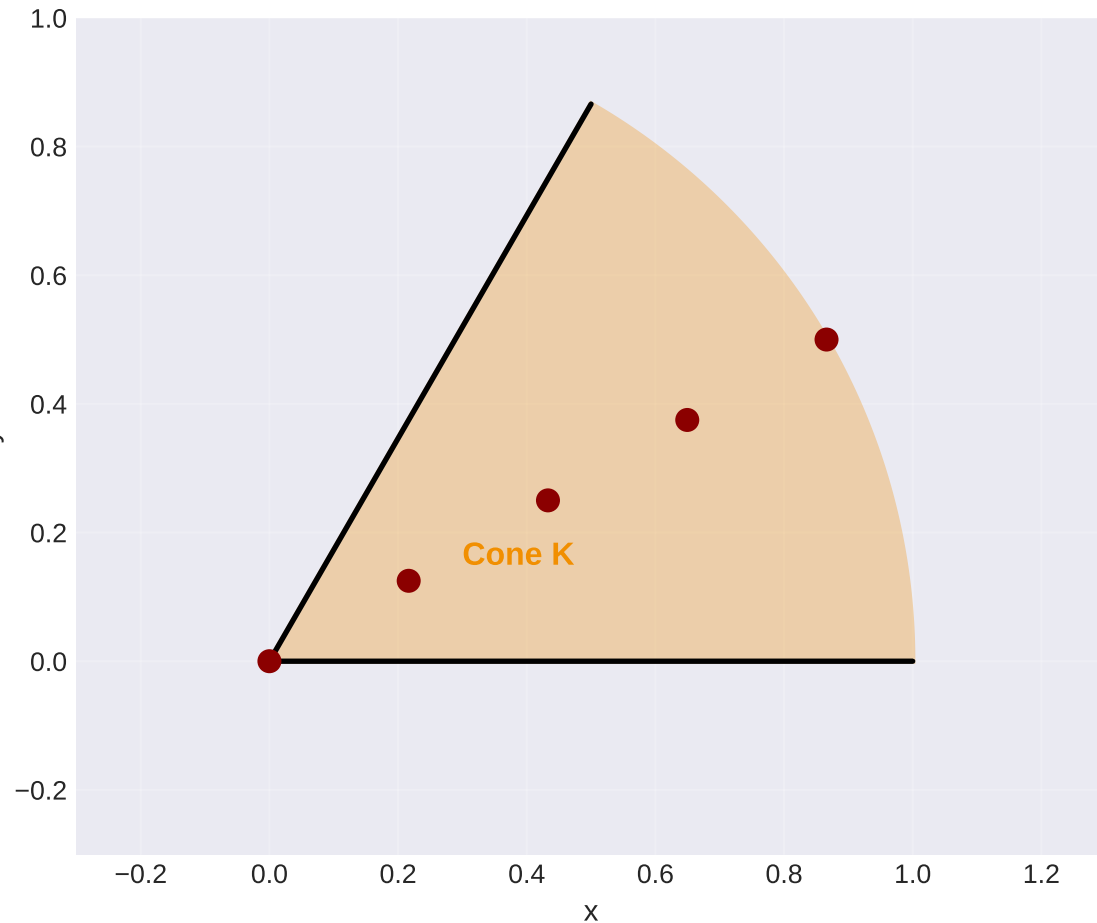


Cone K in  $\mathbb{R}^2$



### Cone Properties (Definition 5.70)

- (i) Closed set
- (ii) Convexity: If  $u, v \in K$ , then  $\alpha u + \beta v \in K$  for all  $\alpha, \beta \geq 0$
- (iii)  $x, -x \in K \iff x = 0$   
(Only origin in both  $K$  and  $-K$ )

### Normal Cone (Definition 5.73)

A cone  $K$  is normal if  $\exists \delta > 0$ :

$$||e_1 + e_2|| \geq \delta, \quad \text{for } e_1, e_2 \in K, \quad ||e_1|| = ||e_2|| = 1$$